Earnings Helper

An automated SEC data extraction, financial metric processing, and synthesis platform delivering actionable intelligence to independent equity traders.

Executive Summary

An independent equities trader managing an active portfolio of 20 to 50 public companies spends over 15 hours during earnings season manually reviewing SEC EDGAR filings, keying financial figures into Excel spreadsheets, and manually calculating year-over-year rate of change metrics. Earnings Helper automates SEC data extraction, financial metric processing, and narrative synthesis into an executive Power BI dashboard, enabling instantaneous portfolio screening and data-driven equity assessment.

The Client Problem & Workflow Bottleneck

- **Manual Data Extraction:** The trader manually accesses SEC 10-Q and 10-K filings for every portfolio ticker across quarterly reporting cycles.
- **Manual Metric Calculation:** Uses calculators and manual Excel formulas to derive year-over-year revenue, gross profit, and operating income percentage growth rates.
- **Scalability Bottleneck:** Managing 20–50 tickers leads to analysis fatigue, delayed trade execution, and high risk of manual human data input errors.
- **Lack of Visual Synthesis:** Financial figures and narrative press release notes remain fragmented across multiple filings, preventing a unified portfolio overview.

Technical Solution Architecture

Ingestion Pipeline & Data Processing

- **SEC Data Engine:** Automated ingestion pipeline utilizing Python and FastAPI to extract SEC financial data directly from the EDGAR APIs.
- **Relational Data Hub:** PostgreSQL database hosted on Supabase storing structured financial statements, historical metrics, and company metadata.
- **AI Analysis Agent:** Automated LLM classification engine evaluating earnings performance into three distinct categories (strong, mixed, weak) and generating structured executive debriefs (headline, takeaways, margin analysis).

Analytics & Business Intelligence Layer

- **PostgreSQL Views:** Standardized relational views delivering pre-calculated growth rates and agent classifications.
- **Power BI Dashboard Integration:** Connected via direct session pooler PostgreSQL connector.
- **Interactive Visualization:** Interactive Donut Chart tracking earnings breakdown, Tile Slicer controls for assessment toggles, Master Company Grid, and dynamic AI Takeaways Multi-row card.

Core Client Deliverables

Deliverable	Description	Key Components
Automated Pipeline Repository	Open-source Python & FastAPI backend	SEC XBRL ingestion engine, scheduled extractors, API router

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Supabase Database Schema	Structured PostgreSQL instance	Relational schema, dynamic rate-of-change view models
Executive Power BI Dashboard	Desktop BI report template & runtime	Interactive cross-filtering, dynamic card views, sentiment trackers

Business Impact & Results

- **15+ Hours Saved:** Reduced quarterly analysis time from over 15 hours down to seconds per earnings cycle.
- **Zero Manual Errors:** Automated data extraction directly from official SEC filings eliminates manual calculator and data entry errors.
- **Actionable Intelligence:** Instant visual segmentation of earnings performance allows immediate focus on underperforming or breakout companies.